

5(a)(i)	<i>any 2 of:</i> 1 to treat disease caused by, faulty / recessive, allele ; A cure 2 deliver, gene / allele / DNA (into target cells of individuals) ; 3 named example ; e.g. SCID / cystic fibrosis	2
5(a)(ii)	(reverse transcriptase) uses mRNA to make (ss)DNA ; (DNA polymerase) makes DNA double-stranded / description ;	2
5(a)(iii)	1 to prevent virus spreading (throughout the body) ; 2 to limit, side effects / immune response / cancer / illness / infection / cell destruction ;	2
5(b)(i)	<i>any 3 of:</i> 1 control ; 2 <i>idea of</i> to see if gene therapy has worked or to compare the effect of no gene therapy with gene therapy or to compare the effect of no hG-CSF gene with presence of hG-CSF gene ; 3 to see where the gene (carried by the vector), goes (in, body / brain) or to see if the gene (carried by the vector), enters cells or to see where cells have been transformed ; 4 to see the effects of the, vector / virus, alone ;	3

Question	Answer	Marks
5(b)(ii)	<i>any 1 of:</i> 1 (control) to compare results (of mice without stroke) to mice with stroke ; 2 to establish baseline figures (in mice without a stroke) / act as a reference point ; 3 to make the study valid ;	1

Question	Answer	Marks						
5(c)	any 4 of: survival 1 increases survival ; brain damage 2 reduces (percentage of brain occupied by) fluid filled space ; behaviour test 3 improves / lowers, behavioural test score ; A score closer to normal general 4 comparative data quote to support, mp1 / mp2 / mp3 ; A processed data <table><tr><td>mp1</td><td>25% v 63% 25% v 100% 63% v 100%</td></tr><tr><td>mp2</td><td>5.2% v 3.6% 5.2% v 3.0% 3.6% v 3.0%</td></tr><tr><td>mp3</td><td>0.90 v 0.67</td></tr></table> 5 eye drops four times is better than eye drops once ; ora 6 if effective with mice likely to be effective with humans ; 7 <i>ref. to</i> unknown differences between mice and humans / study is only on mice ; 8 treatment is, non-invasive / quick ;	mp1	25% v 63% 25% v 100% 63% v 100%	mp2	5.2% v 3.6% 5.2% v 3.0% 3.6% v 3.0%	mp3	0.90 v 0.67	4
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